

Groundworks Playtest

Two different ways of shaping ground are built into the same demo. We want to know which one you'd rather live with when you're clearing a site to put a building on.

Swap models M Step 0.1 m per click Max step 3 m against a neighbour Time needed ~15 min
Build Windows standalone

START HERE

Getting in

1. Unzip the whole folder somewhere and run Terraform.exe. Keep the Terraform_Data folder next to it — the build won't start without it.
2. Windows SmartScreen will warn you about an unsigned app. **More info** → **Run anyway**.
3. You spawn in a flat clearing in the middle of a 64 × 64 m map, one metre per cell. Numbers float on the ground showing heights in metres — on cell middles in the cell model, on the grid corners in the vertex model, because that is where each one keeps its heights.
4. You start in the **cell** model. Press M at any time to swap to the **vertex** model and back.
5. Esc opens a menu with **Resume** and **Quit**.

Each model keeps its own ground. Swapping doesn't reset anything — build something one way, press M, build it the other way, then flip between them to compare. R resets only the model you're currently in.

WHERE TO TRY THINGS

The map

Four bits of ground around the clearing, so you can get straight to a slope instead of spending ten minutes building one. Both models generate the *same* terrain, so anything you compare is the model and not the ground.

MIDDLE

Clearing

Flat, about 20 m across. The empty sandbox — start here, and come back for any test that wants level ground.

NORTH-WEST

One hill

A single clean dome, roughly 5 m tall. The simplest thing to cut a pad into.

NORTH-EAST

Rolling ground

Low and broken, no obvious right answer about where a building goes. The awkward case.

SOUTH-EAST

Small mountain

About 11 m and steep. Still shallow enough per cell that every tool works on it.

SOUTH-WEST

Shallow basin

A dip a couple of metres deep, for testing lowering and backfilling rather than piling.

R puts the whole map back the way it started, in whichever model you're in. It won't touch the other one.

WHAT YOU'RE COMPARING

The two models

MODEL A Cell Every cell has one height, at its middle. You raise a whole cell at a time, and the ground between cells works itself out. Flattening pins that cell level. Level a second pad beside it at a different height and the first gives up its edge, so the two meet in a <i>slope</i> rather than a step.	MODEL B Vertex Heights live at the <i>corners</i>, and that is where the numbers sit. You raise one corner at a time, so a cell tips rather than rises — or raise all four together. Flattening levels the cell between four corners to their <i>average</i> — cutting the high side and filling the low side. Corners are shared, so neighbouring cells move with them.
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REFERENCE

Controls

BOTH MODELS

W A S D · Shift	Walk, sprint. Mouse looks, Space jumps.
M	Swap between the cell and vertex models.
Scroll wheel	Step size per click — 5 cm, 10 cm, 25 cm, 50 cm, 1 m.
Tab	Show or hide the grid lines.
L	Change the light — midday, low sun, overcast. Worth doing before you decide an edge looks wrong.
Z / Y	Undo, redo.
R	Reset the model you're in back to flat.
Esc	Open the pause menu — Resume or Quit. Frees the mouse while it is up.

MODEL A — CELL

- | | |
|-------|--|
| 1 | Sculpt. LMB raises the cell you're looking at, RMB lowers it. Hold to repeat. |
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| 2 | Flatten. LMB levels the cell, and softens any pad touching it — corners included — that was levelled to a different height. It always uses the cell's own current height. |
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| - / = | How far a cell may stand above its neighbour before it refuses — 1, 2, 3, 4 m, or off. |

MODEL B — VERTEX

- | | |
|-------|--|
| 1 | Sculpt a corner. LMB raises the marked corner, RMB lowers it. The dot shows which one. |
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| 4 | Raise a cell. Moves all four corners together, so the cell rises flat instead of tipping. |
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| 2 | Flatten. Levels the outlined cell to the average of its four corners, read fresh each time. |
| <hr/> | |
| 3 | Ramp. LMB at the bottom, LMB at the top, and it grades a slope between. RMB cancels; - = sets its width. |
| <hr/> | |
| - / = | Max step, as above — 0.5, 1, 1.5, 2, 3 m, or off. |

DO THESE IN ORDER • BOTH MODELS EACH TIME

Three tests

01 A hut pad on flat ground

- In the middle clearing, level a patch about three cells square to a single height.
- Do it in the cell model, then press **M** and do it again in the vertex model.
- Level several neighbouring cells and watch what happens where two pads at different heights meet.

Watch for: how many clicks it took, whether the pad came out genuinely flat, and whether you could tell what a click would do *before* you made it.

02 A pad cut into a slope

- Walk out to the **hill** in the north-west and level a three-cell pad into its side — cutting into the high end, filling at the low end.
- Then try the same on the **rolling ground** in the north-east, where there's no obvious flat spot to start from.
- Same again in the other model.

Watch for: what happened to the ground *around* the pad. Did it stay where you left it, or move on its own? Would you be happy putting a wall along that cut edge?

03 Two levels, one metre apart

- On the hill, level a pad, then level the pad right next to it a metre lower.
- Stand at the join and look along it. Then walk down onto the lower one.
- In the vertex model, try **3** to ramp between the two levels.

Watch for: in the cell model the two pads meet in a *slope*, not a step — the older pad gives up its edge rather than leave a stretched vertical face. Does that read better than a hard step would? And does it cost you flat ground you wanted to build on?

Feedback

Answer for building prep specifically — clearing a site to put something on it — not for terrain sculpting in general.

Q1 Which model got you a flat, buildable pad faster? Did faster actually feel better?

Q2 Which was more
predictable where the ground did what you expected before you clicked?

Q3 Which gave you edges you'd want to place a building against?

Q4 In which did the ground you
hadn't touched still look like natural terrain?

Q5 This is a survival game, so groundwork is meant to cost something. Did the effort feel like meaningful work, or like busywork?

Q6 Anything that surprised you, frustrated you, or looked broken?

Please include the cut / fill figure from the panel when you report a test — it's the cubic metres of earth you moved, and it tells us what a pad actually costs in each model.

IGNORE THESE

Known gaps

- There are no textures or props. Judge the *shape* of the ground, not the surface — but do press **L**. Relief is read from shading, and a high midday sun flattens the small steps this test is about, while a low sun makes them obvious.
- The cell model has no ramp tool yet — that's why test 03 only ramps in the vertex model.
- In the vertex model, **T**, **H** and **[]** drive a separate experiment about Unity's built-in terrain. Leave them alone unless you're curious.
- One chunk, no streaming, and the whole map rebuilds on every click. Expect a stutter while you hold a button down — **Tab** hides the grid lines and takes a good chunk of that back. Performance here says nothing about the real game.
- The mountain is there to look at and cut into, not to be levelled flat. That would take a while.